Experiment No: 2

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Subject Name: Full Stack II
Subject Code: 23CSH-309

1. **Aim:** To implement Single Page Application (SPA) navigation in the EcoTrack application using React Router, secure application routes using context-based authentication, and extend nested dashboard routing through follow-up enhancements.

2. **Objective:**

- Configure client-side routing in a React application using React Router
- Implement SPA navigation without full page reloads
- Design and apply protected routes using route-guard patterns
- Manage shared authentication state using React Context API
- Implement nested routing to build dashboard-style layouts
- Extend existing nested routes by adding new dashboard sections
- Implement logout functionality by updating shared context state
- Analyze route access behavior and explain redirection logic
- Understand the role of Context API in shared state management and its comparison with Redux at an introductory level

3. **Implementation / Code:**

```
// App.jsx import { BrowserRouter, Routes, Route } from "react-
router-dom"; import Header from "./components/Header"; import
Login from "./pages/Login";
import DashboardLayout from "./pages/DashboardLayout"; import
DashboardSummary from "./pages/DashboardSummary"; import
DashboardAnalytics from "./pages/DashboardAnalytics"; import
ProtectedRoutes from "./routes/ProtectedRoutes"; function App() {
return (
  <BrowserRouter>
    <Header />
    <Routes>
      <Route path="/login" element={<Login />} />
```

```
        <Route
path="/"
element={
          <ProtectedRoutes>
            <DashboardLayout />
          </ProtectedRoutes>
        }
       >
        <Route index element={<DashboardSummary />} />
        <Route path="summary" element={<DashboardSummary />} />
        <Route path="analytics" element={<DashboardAnalytics />} />
       </Route>
     </Routes>
   </BrowserRouter>
  );
} export default
App;
```

// components/Header.jsx

```
import { Link, useNavigate } from "react-router-dom"; import
{ useAuth } from "../context/AuthContext";

const Header = () => {
  const { isAuth, setIsAuth } = useAuth();
const navigate = useNavigate();

  const handleLogout = () => {
setIsAuth(false);     navigate("/login");
  };

  return (
<div
style={{
        backgroundColor: "#f0f4f8",
padding: "12px 20px",         display:
"flex",         justifyContent: "space-
between",         alignItems: "center",
     }}
   >
```

```
      <h2 style={{ margin: 0 }}>Ecotrack</h2>

      <nav style={{ display: "flex", gap: "12px" }}>
        {isAuth && <Link to="/">Dashboard</Link>}
        {isAuth ? (
          <button onClick={handleLogout}>Logout</button>
        ) : (
          <Link to="/login">Login</Link>
        )}
      </nav>
    </div>
  ); }; export default
Header;
```

// **pages/DashboardLayout.jsx** import { Link,
Outlet } from "react-router-dom"; const
DashboardLayout = () => {

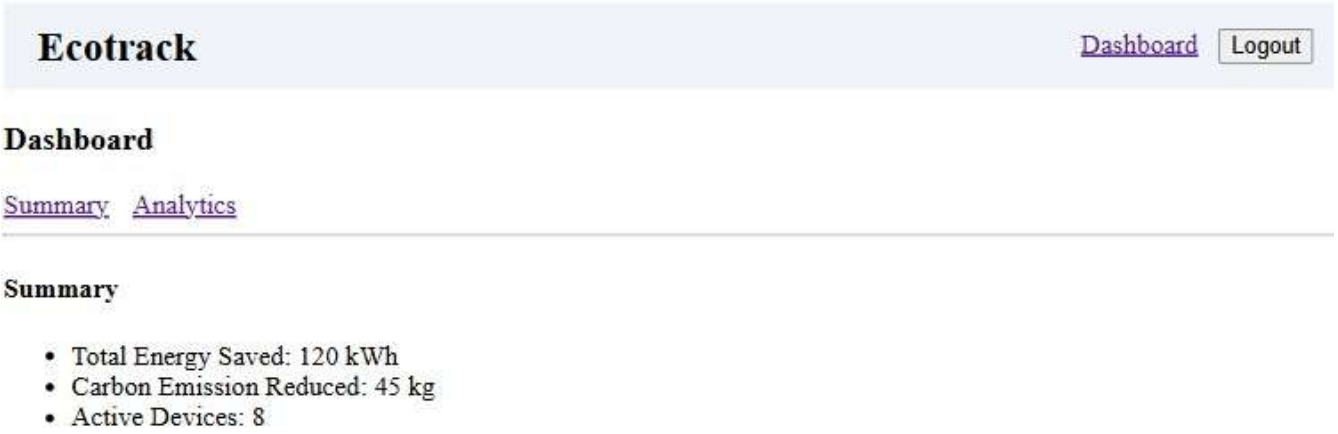
```
  return (
    <div>
      <h3>Dashboard</h3>
      <nav>
        <div style={{display:"flex", gap:"14px"}}>
          <Link to="summary" >Summary</Link>
          <Link to="analytics">Analytics</Link>
        </div>
      </nav>

      <hr />
      <Outlet />
    </div>
  );
};
export default DashboardLayout;
```

// **routes/ProtectedRoutes.jsx**
import { Navigate } from "react-router-dom"; import
{ useAuth } from "../context/AuthContext";

```
const ProtectedRoutes = ({ children }) => {
const { isAuth } = useAuth();   if (!isAuth)
{
   return <Navigate to="/login" replace />;
  }
 return children;
};
export default ProtectedRoutes;
```

4. Output :



5. Learning Outcomes:

- Understood React Router v6 routing (BrowserRouter, Routes, Route, Outlet)
- Learned protected routes using authentication state
- Used Context API for global auth state (AuthContext)
- Implemented login and logout flow
- Conditional rendering based on authentication status
- Used nested routes with a dashboard layout
- Handled programmatic navigation using useNavigate